ARJUN KRISHNAN SURESH

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AI & Data Science undergraduate at Amrita Vishwa Vidyapeetham working across three areas: production LLM systems, empirical deep-learning research, and applied cryptography and network security. Co-author of two research papers; built an offensive-security simulator that implements a man-in-the-middle attack on Diffie-Hellman and the RSA/CA defence that defeats it. Seeking an internship in AI research, model evaluation, or security.

EDUCATION

Amrita Vishwa Vidyapeetham, School of Artificial Intelligence, Faridabad, India **Aug 2024 - May 2028**

B.Tech in Artificial Intelligence & Data Science | CGPA: 8.5 / 10.0

Relevant Coursework: Machine Learning, Deep Learning, Cryptography & Network Security, Computer Networks, Mathematics for Intelligent Systems (convex optimisation, statistical estimation, quantum computing), Data Structures & Algorithms, Database Management Systems, IoT, Robotics

RESEARCH & PUBLICATIONS

Deep Neural Network Driven Intelligent Battery Management System for IoT Nodes **2025**

Published paper, co-author — Python, TensorFlow/Keras, BiLSTM + Multi-Head Attention, GRU, LSTM Autoencoder, Random Forest, SMOTE, scikit-learn

- Benchmarked competing sequence architectures on 19,735 IoT battery sensor readings, split 80/20 chronologically to prevent look-ahead leakage: a 4-head Attention-BiLSTM reached RMSE 1.4776 / MAE 0.4811 / R^2 0.9958 on held-out State-of-Charge prediction, beating the standard BiLSTM baseline (RMSE 1.6141, R^2 0.9950).
- Managed training with Adam (lr 1e-3), batch 64, ≤60 epochs, EarlyStopping (patience 7) and ReduceLROnPlateau; quantified predictive uncertainty via Monte Carlo Dropout, which exposed degraded confidence during charging where the SoC–voltage curve is non-monotonic.
- Reported negative results rather than omitting them: both BiLSTM and GRU battery-health heads returned negative R^2 (−19.29 / −18.74) despite RMSE ≈ 4, and the linear-degradation assumption behind RUL estimation was documented as a stated limitation.
- Engineered 8 feature families (power, thermal stress, rolling, differential, cyclic-time, cycle-depth, interaction, Gaussian-noise augmentation) and added an LSTM autoencoder flagging ~2–3% of windows as anomalies plus a Random Forest + SMOTE operating-mode classifier.

Optimal Emergency Resource Allocation for Urban Disaster Response: Delhi **2026**

Research paper, co-author — MATLAB, Proximal Gradient, ADMM, Dynamic Mode Decomposition, Hankel/SVD, Monte Carlo simulation

- Modelled allocation of scarce relief resources across five Delhi districts — 800 units against 910 units of demand — and benchmarked four optimisers (Proportional, Jacobi, urgency-weighted Proximal Gradient, ADMM); the proximal method scored highest on urgency-weighted fulfilment (U = 0.963 vs 0.951 baseline) while also cutting MSE to 43.17.
- Encoded the welfare trade-off explicitly: a tunable penalty $\lambda = 0.5$ prices the social cost of delaying critical medical supply against routine food distribution, measurably shifting units toward the highest-criticality districts and giving planners a policy dial rather than a fixed rule.
- Validated statistically rather than anecdotally — 200 Monte Carlo runs over Poisson-drawn demand (coefficient of variation < 5%), a 10,000-draw Bernoulli service-probability simulation, and per-region z-scores, p-values and 95% confidence intervals.
- Detected infrastructure failure automatically with Dynamic Mode Decomposition: out-of-unit-circle eigenvalues ($|\lambda_{max}| = 1.14$) isolated blocked routes at Frobenius reconstruction error < 4.8%, and service degraded gracefully (U 0.963 → 0.941) rather than collapsing.

SECURITY & SYSTEMS

Man-in-the-Middle Attack on Diffie-Hellman: Attack & Defence Simulator **2026**

Python (no cryptography libraries), http.server JSON API, HTML/CSS/JS

- Implemented an active man-in-the-middle adversary against unauthenticated Diffie-Hellman — intercepting and substituting both public values, deriving both diverged session keys (g^ac and g^bd), and relaying traffic transparently so the session stays functionally correct and undetectable at the protocol level.
- Built every primitive from scratch instead of calling a library: Miller-Rabin probabilistic primality testing (k = 10 rounds, false-prime probability ≤ 4^{-10}), square-and-multiply modular exponentiation, Extended Euclidean modular inversion, and genuine 512-bit RSA key generation.
- Closed the vulnerability with RSA digital signatures verified through a Certificate Authority, then computed the adversary's forgery attempt explicitly and showed it rejected — demonstrating that Diffie-Hellman is bypassed through an authentication gap, not broken mathematically.
- Analysed real-world failure modes of the deployed protocol: Logjam precomputation against shared 1024-bit parameters, small-subgroup attacks arising from non-primitive generators, and GNFS feasibility below 1024 bits.

Adaptive Bandwidth Monitoring Protocol (ABMP): Anomaly-Based Network Detection **2025**

Python, psutil, NumPy, Streamlit, WebSockets

- Built an anomaly detector over live NIC telemetry that replaces static thresholds with a rolling-window Z-score baseline (N = 20 s, flag at $|Z| \geq 3$), judging traffic against learned volatility — the distinction that separates data exfiltration from a legitimate large download.
- Specified it as an actual protocol: a three-state machine (Calibration → Monitoring → Violation), a JSON Protocol Data Unit carrying ISO-8601 timestamp, throughput, Z-score and status, and a fixed 1-second sampling period.
- Ran passively out-of-band at the application layer against kernel byte counters, adding < 5 ms overhead to the host while flagging spikes to 45.09 MB/s at one-second granularity.

AI & LLM ENGINEERING

CommerceMind VoiceCare AI (Flagship): Voice-First Multilingual Customer Support **2026**

Next.js, FastAPI, LangGraph, Gemini 2.5, Bhashini, ChromaDB, Neon PostgreSQL, Upstash Redis

- Delivered spoken issue resolution in under 10 seconds across 9 Indian languages by architecting a 9-agent LangGraph pipeline (speech-to-text, intent detection, policy RAG, resolution generation, text-to-speech) streamed in real time over WebSocket.
- Proved production readiness with 57 passing unit tests, P50/P95/P99 latency monitoring, and 90% average intent-detection accuracy validated across 35 test conversations/day via a live admin dashboard tracking escalation and resolution analytics.

Production-Ready RAG System: Hybrid Retrieval Engine **2026**

FastAPI, ChromaDB, sentence-transformers, Streamlit, pytest

- Raised retrieval Precision@5 from a 68% cosine-similarity baseline to 84%+ (~24% relative gain) by fusing BM25 keyword search with dense retrieval via Reciprocal Rank Fusion and MS-MARCO cross-encoder reranking, cutting hallucinations by keeping the LLM fed with consistently relevant context.
- Eliminated per-token inference costs via a provider-agnostic LLM_PROVIDER switch between the OpenAI SDK and a local Ollama instance (Llama 3, Mistral), running embeddings and reranking fully offline and validated by a 48-test pytest suite covering deduplication and SSE streaming.

Local LLM Benchmarking & Evaluation Harness **2026**

Python, Ollama, LLM-as-judge, sandboxed code execution

- Benchmarked 6 local LLMs across 40 tasks fully offline under three independent graders — exact match, LLM-as-judge, and live sandboxed code execution — recording per-model latency and throughput so capability and cost could be compared on the same axis.

Natural Language to App Schema Compiler **2026**

Python, Gemini, automated evaluation harness

- Compiled plain-English specifications into validated application schemas through a 5-stage Gemini pipeline, reaching a 90% pass rate across a 20-test automated evaluation harness with a targeted repair engine that retries only the failed stage rather than the whole generation.

Task Manager Agent **2026**

Google ADK, Cloud Run

- Multi-agent productivity assistant coordinating scheduling, note-taking and day planning across specialised sub-agents, deployed as a managed service.

LEADERSHIP & ACHIEVEMENTS

Secretary, Amrita Council of Sports and Excellence **2024 - Present**

- Directed organisation and logistics for 4+ university-wide sports and student-excellence events, coordinating cross-team stakeholders and event operations for 200+ participants.
- Secured ₹10,000 in sponsorship and organised “Respawn Rush,” a campus-wide BGMI esports tournament with 11 teams (44 players), establishing esports as a recognised competitive vertical within the council.

Competitive Athletics **2024 - Present**

- Represented Amrita Vishwa Vidyapeetham at an inter-collegiate badminton tournament in Chennai, finishing 3rd overall; competes across multiple sports for the B.Tech AI & Data Science cohort.

Google Build with AI: Workshop Series **2026**

- Completed three hands-on workshops on agentic AI development (ADK, MCP servers for BigQuery & Google Maps, Gemini-powered applications), organised by Orca AI Club in collaboration with Google.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 AI Foundations Associate, Oracle University (Oct 2025 – Oct 2027) • Google Cloud Gen AI Academy APAC 2026, Cohort 1, Google Cloud × Hack2skill (Apr 2026) • Python Foundation Certification & Artificial Intelligence Primer, Infosys Springboard (Mar 2026)

SKILLS

Languages & Tools: Python, TypeScript/JavaScript, C++, SQL, MATLAB, Git, Docker

AI / ML: TensorFlow, Keras, scikit-learn, NumPy, Pandas, SciPy, BiLSTM / GRU / Multi-Head Attention, LSTM autoencoders, SMOTE, Monte Carlo Dropout, ablation studies

LLMs & Agents: LangGraph, RAG (hybrid retrieval, RRF, cross-encoder reranking), Gemini 2.5, GPT, Ollama (Llama 3, Mistral), Agentic AI (ADK, MCP), LLM-as-judge evaluation harnesses, prompt engineering, vector databases

Security: Applied cryptography (RSA, Diffie-Hellman, Miller-Rabin, modular arithmetic), PKI & certificate authorities, MITM adversary modelling, anomaly-based intrusion detection, TLS/HTTPS, network telemetry

Research & Methods: Convex optimisation (proximal gradient, ADMM, Jacobi), Dynamic Mode Decomposition, Hankel/SVD, Monte Carlo simulation, ablation studies, hypothesis testing & confidence intervals

Backend & Data: FastAPI, Flask, PostgreSQL (Neon), MySQL, Redis (Upstash), SQLAlchemy, ChromaDB, REST APIs, WebSockets, pytest

Frontend & Infra: Next.js, React, Tailwind CSS, Framer Motion, Streamlit, Google Cloud, OCI, Vercel

Spoken Languages: English (Fluent), Hindi (Proficient), Malayalam (Native)